$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^\dagger & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^\dagger{}^{\alpha\beta\chi} & 0 & 3k^2\overset{(4)}{\kappa}_{15}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+\,\alpha\beta} & \mathcal{K}^{\#2}_{1^+\,\alpha\beta} & \mathcal{K}^{\#1}_{1^-\,\alpha} & \mathcal{K}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^\dagger{}^{\alpha\beta} & -\frac{\Upsilon_1 k^2}{6} & \frac{\Upsilon_1 k^2}{3\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^\dagger{}^{\alpha\beta} & \frac{\Upsilon_1 k^2}{3\sqrt{2}} & -\frac{\Upsilon_1 k^2}{3} & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^\dagger{}^{\alpha} & 0 & 0 & -k^2\overset{(4)}{\kappa}_{2} & \frac{k^2\overset{(4)}{\kappa}_{2}}{\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}{}^\dagger{}^{\alpha} & 0 & 0 & \frac{k^2\overset{(4)}{\kappa}_{2}}{\sqrt{2}} & -\frac{k^2\overset{(4)}{\kappa}_{2}}{2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+\,\alpha\beta} & \mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^\dagger{}^{\alpha\beta} & \frac{3\Upsilon_1 k^2}{2} & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^\dagger{}^{\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^\dagger & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^\dagger{}^{\alpha\beta\chi} & 0 & \frac{1}{3k^2\overset{(4)}{\kappa}_{15}}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+\,\alpha\beta} & \mathcal{J}^{\#2}_{1^+\,\alpha\beta} & \mathcal{J}^{\#1}_{1^-\,\alpha} & \mathcal{J}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^\dagger{}^{\alpha\beta} & \frac{1}{3\,\mathrm{Det}(1^+)} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^\dagger{}^{\alpha\beta} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & \frac{2}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^\dagger{}^{\alpha} & 0 & 0 & -\frac{4}{9k^2\overset{(4)}{\kappa}_{2}} & \frac{2\sqrt{2}}{9k^2\overset{(4)}{\kappa}_{2}} \\
\mathcal{J}^{\#2}_{1^-}{}^\dagger{}^{\alpha} & 0 & 0 & \frac{2\sqrt{2}}{9k^2\overset{(4)}{\kappa}_{2}} & -\frac{2}{9k^2\overset{(4)}{\kappa}_{2}}
\end{array}
$$

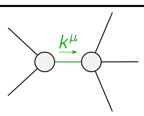
$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+\,\alpha\beta} & \mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^\dagger{}^{\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^\dagger{}^{\alpha\beta\chi} & 0 & 0
\end{array}
$$

Abbreviations used in matrices
$\Upsilon_1 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2 \;\&\&\; \mathrm{Det}(1^+) == -\frac{1}{2}k^2(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2) \;\&\&\; \mathrm{Det}(2^+) == \frac{3}{2}k^2(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2)$

Lagrangian
$\overset{(4)}{\kappa}_1\partial_\beta\mathcal{K}^\delta{}_{\chi\delta}\partial^\chi\mathcal{K}^\alpha{}_\alpha{}^\beta + \overset{(4)}{\kappa}_2\partial_\chi\mathcal{K}^\delta{}_{\beta\delta}\partial^\chi\mathcal{K}^\alpha{}_\alpha{}^\beta - \frac{4}{3}\overset{(4)}{\kappa}_1\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\alpha\chi}{}^\delta + 2\overset{(4)}{\kappa}_{15}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\alpha\chi}{}^\delta -$ $\frac{4}{3}\overset{(4)}{\kappa}_2\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\alpha\chi}{}^\delta + \frac{2}{3}\overset{(4)}{\kappa}_1\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^\delta - 4\overset{(4)}{\kappa}_{15}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^\delta + \frac{2}{3}\overset{(4)}{\kappa}_2\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}{}^\delta -$ $\frac{5}{3}\overset{(4)}{\kappa}_1\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^\delta - 2\overset{(4)}{\kappa}_{15}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^\delta - \frac{5}{3}\overset{(4)}{\kappa}_2\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}{}^\delta - \frac{1}{3}\overset{(4)}{\kappa}_1\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} -$ $\overset{(4)}{\kappa}_{15}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} - \frac{1}{3}\overset{(4)}{\kappa}_2\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\partial_\delta\mathcal{K}_{\alpha\beta\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi} - 2\overset{(4)}{\kappa}_{15}\partial_\delta\mathcal{K}_{\beta\alpha\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi}$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$

Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta\mathcal{J}^\alpha{}_\alpha{}^\beta == 0$
$\mathcal{J}^{\#1\,\alpha}_{1^-} + 2\,\mathcal{J}^{\#2\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_\beta == 3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$2\,\mathcal{J}^{\#1\,\alpha\beta}_{1^+} + \mathcal{J}^{\#2\,\alpha\beta}_{1^+} == 0$	3	$\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\chi\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_\delta + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} +$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^\delta{}_\delta{}^\epsilon + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_\delta ==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_\delta + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} +$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^\delta{}_\delta{}^\epsilon + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_\delta$
Total # constraint(s):	12	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{\overset{(4)}{\kappa}_2}$

Resolved unitarity condition(s):	$\overset{(4)}{\kappa}_2 > 0$